

Notebook on
Numerical Methods

Prepared by
Shahriar Arefin

ID: 22101040

Section: (A)

Faculty
Dr. Homeyra Akter

Semester: Fall 2024

Contents

1	Concepts	2
---	----------	---

Continuous Functions

$$\therefore \text{Forward Difference Approximation} \quad f'(x) = \frac{f(x+h) - f(x)}{h} = \frac{f(x_{i+1}) - f(x_i)}{\Delta x}$$

$$\therefore \text{Backward Difference Approximation} \quad f'(x) = \frac{f(x) - f(x-h)}{h} = \frac{f(x_i) - f(x_{i-1})}{\Delta x}$$

$$\therefore \text{Central Difference Approximation} \quad f'(x) = \frac{f(x+h) - f(x-h)}{2h} = \frac{f(x_{i+1}) - f(x_{i-1})}{2\Delta x}$$

$$\therefore \text{Absolute relative true error, } |\epsilon_t| = \left| \frac{\text{True value} - \text{Approximate value}}{\text{True value}} \right| \times 100$$

Bisection Method

STEP-1

Assume x_l and x_u . Choose x_l and x_u as two guesses for the root such that $f(x_l)f(x_u) < 0$, or in other words, $f(x)$ changes sign between x_l and x_u .

STEP-2

Estimate the root, x_m of the equation $f(x) = 0$ as the mid point between x_l and x_u as $x_m = \frac{x_l + x_u}{2}$.

STEP-3

1. If $f(x_l)f(x_m) < 0$, then the root lies between x_l and x_m ; then $x_l = x_l$ and $x_u = x_m$.
2. If $f(x_l)f(x_m) > 0$, then the root lies between x_l and x_m ; then $x_l = x_m$ and $x_u = x_u$.
3. If $f(x_l)f(x_m) = 0$, then the root is x_m ; Stop the algorithm if this is true.

STEP-4

Find the new estimate of the root, $x_m = \frac{x_l + x_u}{2}$

Find the absolute relative approximate error, $|\epsilon_t| = \left| \frac{x_m^{\text{new}} - x_m^{\text{old}}}{x_m^{\text{new}}} \right| \times 100$

STEP-5

Compare the absolute relative approximate error with the pre-specified error tolerance- $|\epsilon_a| \leq 0.5 \times 10^{2-m}\%$

- If $m = 1$, then $|\epsilon| \leq 5\%$
- If $m = 2$, then $|\epsilon| \leq 0.5\%$
- If $m = 3$, then $|\epsilon| \leq 0.05\%$

Chapter 1

Concepts

Continuous Functions

Forward Difference

According to Taylor's theorem,

$$\begin{aligned} f(x_{i+1}) &= f(x_i) + f'(x_i)(\Delta x) + \frac{f''(x_i)}{2!}(\Delta x)^2 + \dots \quad [\text{where } x = x_{i+1} - x_i] \\ \Rightarrow f'(x_i) &= \frac{f(x_{i+1}) - f(x_i)}{\Delta x} - \frac{f''(x_i)}{2!}(\Delta x) + \dots \\ \Rightarrow f'(x_i) &= \frac{f(x_{i+1}) - f(x_i)}{\Delta x} + 0(\Delta x) \quad [\text{Neglecting higher-order terms}] \end{aligned}$$

Backward Difference

According to Taylor's theorem,

$$\begin{aligned} f(x_{i-1}) &= f(x_i) - f'(x_i)(\Delta x) + \frac{f''(x_i)}{2!}(\Delta x)^2 + \dots \quad [\text{where } x = x_i - x_{i-1}] \\ \Rightarrow f'(x_i) &= \frac{f(x_i) - f(x_{i-1})}{\Delta x} + \frac{f''(x_i)}{2!}(\Delta x) + \dots \\ \Rightarrow f'(x_i) &= \frac{f(x_i) - f(x_{i-1})}{\Delta x} + 0(\Delta x) \quad [\text{Neglecting higher-order terms}] \end{aligned}$$

Central Difference

According to Taylor's theorem,

$$f(x_{i+1}) = f(x_i) + f'(x_i)(\Delta x) + \frac{f''(x_i)}{2!}(\Delta x)^2 + \frac{f'''(x_i)}{3!}(\Delta x)^3 + \dots \quad [\text{where } x = x_{i+1} - x_i] \quad (1)$$

$$f(x_{i-1}) = f(x_i) - f'(x_i)(\Delta x) + \frac{f''(x_i)}{2!}(\Delta x)^2 - \frac{f'''(x_i)}{3!}(\Delta x)^3 + \dots \quad [\text{where } x = x_i - x_{i-1}] \quad (2)$$

Subtracting equation (2) from equation (1)

$$\begin{aligned} \Rightarrow f(x_{i+1}) - f(x_{i-1}) &= 2f'(x_i)(\Delta x) + \frac{2f'''(x_i)}{3!}(\Delta x)^3 + \dots \\ \Rightarrow \frac{f(x_{i+1}) - f(x_{i-1})}{2(\Delta x)} &= f'(x_i) + \frac{f'''(x_i)}{3!}(\Delta x)^2 + \dots \\ \Rightarrow f'(x_i) &= \frac{f(x_{i+1}) - f(x_{i-1})}{2(\Delta x)} - \frac{f'''(x_i)}{3!}(\Delta x)^2 + \dots \\ \Rightarrow f'(x_i) &= \frac{f(x_{i+1}) - f(x_{i-1})}{2(\Delta x)} - 0(\Delta x)^2 \quad [\text{Neglecting higher-order terms}] \end{aligned}$$

Newton-Raphson

Step-1 : Calculate $f'(x)$ symbolically.

Step-2 : Choose an initial guess of the root x_i , to estimate the new value of the root x_{i+1} as,

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Step-3 : Find the absolute relative approximate error $|\epsilon_a|$ as

$$|\epsilon_a| = \left| \frac{x_{i+1} - x_i}{x_{i+1}} \right| \times 100$$

Step-4 : Compare the absolute relative approximate error with the pre-specified relative error tolerance $|\epsilon_s|$.

Step-5 : If $|\epsilon_a| \geq |\epsilon_s|$, then Go to Step 2 using new estimate of the root.
Else stop the algorithm.

Secant Method

Step-1 : Calculate the next estimate of the root from two initial guesses x_i and x_{i-1} ,

$$x_{i+1} = x_i - \frac{f(x_i)(x_i - x_{i-1})}{f(x_i) - f(x_{i-1})}$$

Step-2 : Find the absolute relative approximate error $|\epsilon_a|$ as

$$|\epsilon_a| = \left| \frac{x_{i+1} - x_i}{x_{i+1}} \right| \times 100$$

Step-3 : Compare the absolute relative approximate error with the pre-specified relative error tolerance $|\epsilon_s|$.

Step-4 : If $|\epsilon_a| \geq |\epsilon_s|$, then Go to Step 1 using new estimate of the root.
Else stop the algorithm.

Gaussian Elimination

Forward Elimination :

- Transform the system into an upper triangular matrix using elementary row operations.
- To simplify the system so each equation has fewer variables/unknowns than the one above.

Back Substitution :

- Solve the simplified upper triangular system from the bottom row upward.
- To find the values of the variables/unknowns in the original system of equations.

LU Decomposition

Given the system of equations $[A][X] = [C]$, where:

- $[A]$ is the coefficient matrix;
- $[X]$ is the unknown matrix;
- $[C]$ is the constant matrix;

Decompose $[A]$ into lower and upper triangular matrices: Such that $[A] = [L][U]$. Where:

- $[U]$: upper triangular matrix; obtained by applying elementary row operations on $[A]$. (forward elimination)
- $[L]$: lower triangular matrix; obtained by the multipliers used in the forward elimination steps.

Forward substitution: Solve $[L][Z] = [C]$ for $[Z]$

Back substitution: Solve $[U][X] = [Z]$ for $[X]$

The solution $[X]$ gives the values of the unknowns in the original system.

Trapezoidal Rule

Single Segment :

$$\begin{aligned}
 \int_a^b f(x)dx &\approx \text{Area of trapezoid} \\
 &= \frac{1}{2}(\text{Sum of parallel sides})(\text{height}) \\
 &= \frac{1}{2}[f(b) + f(a)](b - a) \\
 &= \frac{(b - a)}{2}(f(a) + f(b))
 \end{aligned}$$

Multiple Segment :

$$\begin{aligned}
 \int_a^b f(x)dx &\approx \frac{h}{2}[f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_{n-1}) + f(x_n)] \quad \text{where, } h = \frac{b - a}{n} \\
 &= \frac{h}{2} \left[f(x_0) + 2 \left\{ \sum_{i=1}^{n-1} f(x_i) \right\} + f(x_n) \right] \\
 &= \frac{b - a}{2n} \left[f(a) + 2 \left\{ \sum_{i=1}^{n-1} f(a + ih) \right\} + f(b) \right]
 \end{aligned}$$

Simpson's 1/3rd Rule

Multiple Segment :

$$\begin{aligned}
 \int_a^b f(x)dx &= \frac{h}{3}[f(x_0) + 4[f(x_1) + f(x_3) + \dots + f(x_{n-1})] + 2[f(x_2) + f(x_4) + \dots + f(x_{n-2})] + f(x_n)] \\
 &= \frac{h}{3} \left[f(x_0) + 4 \left\{ \sum_{\substack{i=1 \\ i \text{ odd}}}^{n-1} f(x_i) \right\} + 2 \left\{ \sum_{\substack{i=2 \\ i \text{ even}}}^{n-2} f(x_i) \right\} + f(x_n) \right] \\
 &= \frac{b - a}{3n} \left[f(a) + 4 \left\{ \sum_{\substack{i=1 \\ i \text{ odd}}}^{n-1} f(a + ih) \right\} + 2 \left\{ \sum_{\substack{i=2 \\ i \text{ even}}}^{n-2} f(a + ih) \right\} + f(b) \right]
 \end{aligned}$$

Newton Divided Difference

Quadratic :

Given (x_0, y_0) , (x_1, y_1) , and (x_2, y_2) , fit a quadratic interpolant through the data. $f_2(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1)$

$$b_0 = f(x_0)$$

$$b_1 = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$b_2 = \frac{\frac{f(x_2) - f(x_1)}{x_2 - x_1} - \frac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0}$$

Euler's Method

Use Euler's method with $h=0.1$ to find approximate values for the solution of the initial value problem

$$\frac{dy}{dx} + 2y = x^3 e^{-2x}, y(0) = 1 \text{ at } x=0.1, 0.2, 0.3$$

Solution:

Given,

$$\frac{dy}{dx} + 2y = x^3 e^{-2x}, y(0) = 1$$

Rewriting the equation in differential form,

$$\frac{dy}{dx} = x^3 e^{-2x} - 2y$$

From given equation, $h = 0.1$ $x_0 = 0$, $y_0 = 1$

As we know, $x_{i+1} = x_i + h$

$$x_1 = x_0 + h = 0 + 0.1 = 0.1$$

Now, $x_2 = x_1 + h = 0.1 + 0.1 = 0.2$

$$x_3 = x_2 + h = 0.2 + 0.1 = 0.3$$

$$y_1 = y_0 + f(x_0, y_0) \times h$$

$$= 1 + f(0, 1) \times 0.1$$

$$= 0.8$$

$$y_2 = y_1 + f(x_1, y_1) \times h$$

$$= 0.8 + f(0.1, 0.8) \times 0.1$$

$$= 0.640081$$

$$y_3 = y_2 + f(x_2, y_2) \times h$$

$$= 0.640081 + f(0.2, 0.640081) \times 0.1$$

$$= 0.5126017$$

Gaussian Elimination

Question: Solve the system of linear equations using Naive Gaussian Elimination,

$$25x^3 + 5x^2 + x = 106.8$$

$$64x^3 + 8x^2 + x = 177.2$$

$$144x^3 + 12x^2 + x = 279.2$$

Solution:

Writing the system of linear equations in $[A][x] = [C]$ form,

Now,

$$\left(\begin{array}{ccc|c} 25 & 5 & 1 & 106.8 \\ 64 & 8 & 1 & 177.2 \\ 144 & 12 & 1 & 279.2 \end{array} \right)$$

$$\Rightarrow \left(\begin{array}{ccc|c} 25 & 5 & 1 & 106.8 \\ 0 & -4.8 & -1.56 & -96.208 \\ 0 & -16.8 & -4.76 & -335.968 \end{array} \right) \begin{array}{l} R_2 = R_2 - \frac{64}{25}R_1 \\ R_3 = R_3 - \frac{144}{25}R_1 \end{array}$$

$$\Rightarrow \left(\begin{array}{ccc|c} 25 & 5 & 1 & 106.8 \\ 0 & -4.8 & -1.56 & -96.208 \\ 0 & 0 & 0.7 & 0.76 \end{array} \right) R_3 = R_3 - \frac{-16.8}{-4.8}R_2$$

$$\Rightarrow \left(\begin{array}{ccc|c} 1 & 0.2 & 0.04 & 4.272 \\ 0 & 1 & 0.325 & 20.043 \\ 0 & 0 & 1 & 1.0857 \end{array} \right) \begin{array}{l} R_1 = R_1 \div 25 \\ R_2 = R_2 \div -4.8 \\ R_3 = R_3 \div 0.7 \end{array}$$

Expressing this in equations form,

$$x_1 + 0.2x_2 + 0.04x_3 = 4.272$$

$$x_2 + 0.325x_3 = 20.043$$

$$x_3 = 1.0857$$

By back substitution,

$$\therefore x_3 = 1.0857$$

$$\therefore x_2 = 20.043 - 0.325x_3$$

$$= 20.043 - 0.325(1.0857) = 19.6901$$

$$\therefore x_1 = 4.272 - 0.04x_3 - 0.2x_2$$

$$= 4.272 - 0.04(1.0857) - 0.2(19.6901) = 0.2906$$

Newton-Raphson

Question: Use the Newton-Raphson method to find the root of the equation $x^3 - 0.165x^2 + 3.993 \times 10^{-4} = 0$.
Use 0.05 as the initial guess.

Solution:

$$\text{Let } f(x) = x^3 - 0.165x^2 + 3.993 \times 10^{-4}$$

$$\text{Then, } f'(x) = 3x^2 - 0.33x$$

$$\text{Let, } x_0 = 0.05$$

Iteration-1:

$$\begin{aligned} x_1 &= x_0 - \frac{f(x_0)}{f'(x_0)} \\ &= 0.05 - \frac{f(0.05)}{f'(0.05)} \\ &= 0.05 - \frac{(0.05)^3 - 0.165(0.05)^2 + 3.993 \times 10^{-4}}{3(0.05)^2 - 0.33(0.05)} \\ &= 0.06242 \end{aligned}$$

Iteration-2:

$$\begin{aligned} x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\ &= 0.06242 - \frac{(0.06242)^3 - 0.165(0.06242)^2 + 3.993 \times 10^{-4}}{3(0.06242)^2 - 0.33(0.06242)} \\ &= 0.06238 \end{aligned}$$

$$\therefore |\epsilon_a| = \left| \frac{0.06238 - 0.06242}{0.06238} \right| \times 100 = 0.0716\%$$

Iteration-3:

$$\begin{aligned} x_3 &= x_2 - \frac{f(x_2)}{f'(x_2)} \\ &= 0.06238 - \frac{(0.06238)^3 - 0.165(0.06238)^2 + 3.993 \times 10^{-4}}{3(0.06238)^2 - 0.33(0.06238)} \\ &= \mathbf{0.06238} \end{aligned}$$

$$\therefore |\epsilon_a| = \left| \frac{0.06238 - 0.06238}{0.06238} \right| \times 100 = 0\%$$

Secant Method

Question: Use the Secant method to find the root of the equation $x^3 - 0.165x^2 + 3.993 \times 10^{-4} = 0$.
Use 0.02 and 0.055 as the initial guesses.

Solution:

$$\text{Let } f(x) = x^3 - 0.165x^2 + 3.993 \times 10^{-4}$$

$$\text{and } x_{-1} = 0.02, \quad x_0 = 0.05$$

Iteration-1:

$$\begin{aligned} x_1 &= x_0 - \frac{f(x_0)(x_0 - x_{-1})}{f(x_0) - f(x_{-1})} \\ &= 0.05 - \frac{f(0.05)(0.05 - 0.02)}{f(0.05) - f(0.02)} \\ &= 0.06242 \end{aligned}$$

Iteration-2:

$$\begin{aligned} x_2 &= x_1 - \frac{f(x_1)}{f'(x_1)} \\ &= 0.06242 - \frac{(0.06242)^3 - 0.165(0.06242)^2 + 3.993 \times 10^{-4}}{3(0.06242)^2 - 0.33(0.06242)} \\ &= 0.06238 \end{aligned}$$

$$\therefore |\epsilon_a| = \left| \frac{0.06238 - 0.06242}{0.06238} \right| \times 100 = 0.0716\%$$

Iteration-3:

$$\begin{aligned} x_3 &= x_2 - \frac{f(x_2)}{f'(x_2)} \\ &= 0.06238 - \frac{(0.06238)^3 - 0.165(0.06238)^2 + 3.993 \times 10^{-4}}{3(0.06238)^2 - 0.33(0.06238)} \\ &= \mathbf{0.06238} \end{aligned}$$

$$\therefore |\epsilon_a| = \left| \frac{0.06238 - 0.06238}{0.06238} \right| \times 100 = 0\%$$

Trapezoidal Rule

$$h = \frac{b-a}{n} = \frac{30-8}{4} = 5.5$$

$$\begin{aligned} & \frac{b-a}{2n} \left[f(a) + 2 \left\{ \sum_{i=1}^{n-1} f(a+ih) \right\} + f(b) \right] \\ &= \frac{30-8}{2 \times 4} \left[f(a) + 2 \left\{ \sum_{i=1}^{4-1} f(a+ih) \right\} + f(b) \right] \\ &= \frac{30-8}{2 \times 4} [f(a) + 2f(a+h) + 2f(a+2h) + 2f(a+3h) + f(b)] \\ &= \frac{30-8}{2 \times 4} [f(8) + 2f(8+5.5) + 2f(8+11) + 2f(8+16.5) + f(30)] \end{aligned}$$